Multi-Sensor Data Fusion, AIoT, and Intelligent Modeling for Sustainable Smart Cities

Kim Sarah Wurster (a24kimwu)
University of Skövde, Sweden
Course: Information Fusion
a24kimwu@student.his.se

December 21, 2025

1 Introduction

The rapid urbanization and increasing complexity of urban systems (Bibri & Huang, 2025b; Jadhav et al., 2024) result in a need for data-driven, Artificial Intelligence (AI)-enhanced tools to enable sustainable smart cities. These are needed to ensure mobility, energy efficiency and optimal decision-making across all stages and domains of urban city planning. Multi-sensor fusion, AIoT (AI + Internet of Things) and deep learning form the basis of modern smart city infrastructures and new framework ideas.

Urban planners need predictive systems that can integrate heterogeneous data sources and provide information in real-time. Existing systems face challenges in data heterogeneity and uncertainty (Ma, 2025), real-time decision-making (Gantla et al., 2025) as well as energy efficiency and cybersecurity (Alssaiari et al., 2025; Nassra & Capella, 2025). The aim of this literature review is to analyze and evaluate the current research on multi-sensor fusion frameworks and AIoT systems for sustainable smart city development. The goal is to identify recurring themes and research gaps that can be a direction for future work.

The review is organized in four themes that occurred in the literature: Spatiotemporal and multisource data fusion for sustainable urban planning,Sensor fusion for environmental monitoring and urban sustainability, AIoT, CPSoS, and intelligent urban governance, Energy efficiency and cybersecurity. Within each of these themes, the included papers are summarized, analyzed and critically evaluated based on the two focuses: information fusion and the handling of uncertainty.

2 Themes

2.1 Spatiotemporal & Multisource Data Fusion for Sustainable Urban Planning

Spatiotemporal and multisource data fusion is an important part of sustainable urban planning since it enables cities to integrate heterogeneous data streams to create valuable insights for improved decision-making. Recent research shows how advanced machine learning (ML), probabilistic modeling, and intelligent Internet of Things (IoT) integration can improve sustainable urban planning.

Ma (2025) presents an innovative multi-source information fusion framework called the Adaptive Bayesian Hierarchical Multi-Source Fusion (ABHMF) system which can be seen in Figure 1. This system has multiple different inputs: satellite images, ground weather stations, IoT sensors, urban morphology datasets and numerical weather predictions. The author combines three stages to fuse the data: data preprocessing with gap-filling and error checking to ensure a high quality, spatiotemporal alignment to integrate data from different resolutions and frequencies and the adaptive Bayesian hierarchical fusion. In the adaptive Bayesian hierarchical fusion they combine adaptive deep learning for dynamic weight adjustments with a Bayesian hierarchical architecture that quantifies and decomposes uncertainty. The framework provides outputs like temperature, humidity and wind together with uncertainty metrics and reports a strong performance with an RMSE of 0.51 °C and a fast runtime of around 10 s per daily forecast.

Uncertainty is an integrated element in Ma (2025)'s paper. He uses the law of total variance to quantify the total uncertainty adding the expectation of conditional variance representing aleatoric components which have uncertainty inherent in the data and the variance of conditional expectations representing epistemic components which have uncertainty from parameter estimation. Through statistics using normal distributions for aggregated sensor errors, variational inference and the three-cornered hat method, the model adaptively weighs each incoming data source based on real-time quality, reducing the prediction error and maintaining computational efficiency. This probabilistic fusion approach allows the model to produce climate estimates as well as insightful uncertainty metrics, making the framework suitable for risk-sensitive urban climate planning.
The paper integrates realistic data sources and showcases a practical orientation which makes it applicable for the real world. The explicit uncertainty propagation and decomposition is valuable to support decision-making. One disadvantage is that IoT sensors vary widely in quality and although the adaptive weights help, does the process for robustly detecting errors need to be further investigated.

In their paper Jadhav et al. (2024) propose a generative adversarial framework that fuses satellite images, census data, land-use maps, transportation data, and data from environmental sensors to synthesize spatiotemporal urban data. This data undergoes a spatiotemporal fusion process resulting in pollution maps that are integrated into a database with the original data. This spatiotemporal fusion process is done using statistical methods and ML algorithms that are fine-tuned for the specific spatial context and temporal dynamics of a specific geographical layout. Information fusion is further enhanced through a Generative Adversarial Network (GAN) structure in which synthetic urban patterns are generated and validated through an adversarial process. The adversarial training iterations help produce realistic simulations. The GAN outputs high-resolution urban growth forecasts and scenario simulations (2025–2050) that could be fed into an agent-based model representing policy and society to simulate the response to new circumstances according to Jadhav et al. (2024). They also test classifiers and show that multimodal fusion improves classification and planning insights. The cities Bangalore, Detroit and Shenzhen were used as test cities to showcase different applications and the scenarios were evaluated under different governmental interventions.
While uncertainty is not modeled probabilistically as in Ma (2025), it is handled indirectly through the simulations. The repeated GAN simulations provide a loophole for uncertainty-aware planning through allowing the model to explore multiple potential trajectories of urban expansion, providing planners with an understanding of long-term urban growth outcomes.
The use of simulations gives stakeholders the chance to see realistic outcomes and try ideas without risk which is a great advantage for urban planning. A disadvantage is the usage of census and social media data which often have representation biases which can affect the GAN outputs. Another point is that GANs are black box generators and planners often need interpretable drivers therefore adding explainability would be a great improvement of the framework.

Chithaluru et al. (2023) focus on neuro-fuzzy techniques for IoT networks to achieve dynamic clustering and routing to improve lifetime, throughput and energy efficiency. Their neuro-fuzzy clustering solution selects cluster heads (CHs) based on node features like energy availability, mobility, storage, hops, and bandwidth using a Gradient-based Feed-Forward Neural Network

(GFF-NN) combined with fuzzy logic. The Gradient-based Feed-Forward Neural Network then fuses node attributes into cluster assignments, helping determine optimal cluster heads. These CHs collect information from their group nodes and transfer it to a distant sink. This fusion approach enables dynamic load balancing and optimized routing. The evaluation is done based on simulations and the proposed network's life duration is improved by 50-60% when being compared to the low-energy adaptive clustering hierarchy with reduced overhead due to dynamic clustering. While their work does not model predictive uncertainty, it addresses uncertainty in input conditions by using fuzzy logic and the GFF-NN to evaluate node reliability. The approach acknowledges IoT constraints and uses clustering to reduce communication and extend lifetime. The paper from Chithaluru et al. (2023) addresses the common problem of energy efficiency with sensor networks which gives it a practical orientation and the routing improvements directly reduce energy consumption by optimising which nodes act as CHs. One weakness is that the simulation outcomes are promising but depending on the simulation assumptions the real world performance might be different as the simulations may not accurately replicate the real deployment challenges.

All of these papers have multimodal data fusion as a central topic and use ML as an enabler. Ma (2025) explicitly model uncertainty, Jadhav et al. (2024) use simulations and Chithaluru et al. (2023) address uncertainty via fuzzy logic. Combining statistics with ML is a common trajectory in these papers where the authors blend strengths of the two.
Chithaluru et al. (2023)'s neuro-fuzzy approach can be used to strengthen the IoT data quality and uptime of Ma (2025)'s ABHMF. This would lead to better routing and clustering due to fewer missing data and lower measurement uncertainty. Jadhav et al. (2024)'s GAN-based scenarios can be fed into ABHMF or similar models to assess climate impacts under alternative urban futures.

2.2 Sensor Fusion for Environmental Monitoring and Urban Sustainability

Sensor fusion plays a critical role in environmental monitoring and urban sustainability by integrating heterogeneous sensor streams to produce accurate, timely, and actionable insights. Recent research shows a clear shift toward hybrid, edge-enabled fusion architectures that prioritize robustness, scalability, and sustainability.

Reis (2025) focuses on traffic optimization through a multimodal data fusion framework using LSTM-based forecasting for congestion prediction and Deep Q-Network (DQN) for route optimization. The approach uses multimodal data fusion across EV sensors, traffic systems and environmental datasets through AI models using learnable weights which improve real-time decisions in route selection and energy efficiency. Fusion occurs at the preprocessing stage, where heterogeneous inputs are synchronized and passed to the LSTM and DQN this improves forecasting accuracy and routing effectiveness. The travel time was reduced by 20% and 15% energy savings were achieved. The approach was tested in a simulation environment that imitates a scalable edge-cloud hybrid architecture with urban mobility simulation, IoT device emulation, an edge computing layer, a cloud processing layer and edge-cloud coordination to ensure real-time predictions.
One weakness of this approach is that the heavy reliance on simulation limits real-world trans-

ferability and additionally to that the framework does not include uncertainty awareness for the sensors or a detailed explanation about the data fusion in the approach.

In their paper Mohanty et al. (2025) focus on environmental monitoring and evaluate two fusion strategies: weighted averaging and support vector machine (SVM) classification. The goal is to combine air quality, temperature, humidity, noise, and wind sensor data. Weighted averaging means that every sensor output is normalized and added to the single output stream with its own weight. This produces stable, low-error outputs suitable for real-time applications. The weighted average fusion method is fast, reliable and has a low computational cost. In the SVM-based approach a SVM is used to train historical sensor data with higher accuracy in the predictions. "It uses a kernel function to map inputs into a higher-dimensional space for better feature separability" (Mohanty et al., 2025). The SVM based approach is more accurate and adaptive but also computationally more expensive. For environmental prediction tasks the SVM is a great fit, the research paper showed that ML models outperform statistical methods. The method does not handle uncertainty by itself, since the sensor outputs are pre-processed to handle inconsistent or noisy data using data cleaning and normalizing so the sensor fusion models only receive only the best inputs. In order to use this method in a real life application the preprocessing needs to be included in the strategies and uncertainty awareness needs to be implemented as well.

Gantla et al. (2025) propose a comprehensive multi-modal fusion system combining traffic cameras, GPS tracks, loop detectors, LiDAR, and environmental monitors. Their hybrid AI fusion architecture is hierarchical and includes three stages which can be seen in Figure 2: data preprocessing with Kalman filtering and PCA, feature extraction with the help of a CNN for spatial features from cameras, a Random Forest Regression (RFR) for pollution prediction and a LSTM for temporal patterns prediction and adaptive control via reinforcement learning (RL) for decision-making. This results in an advanced fusion pipeline, combining preprocessing, feature-level fusion and decision-level RL for signal control and traffic diversion. These actions are executed at the Edge AI layer to ensure real-time urban operations. The fusion of traffic and pollution data enables pollution-aware routing and adaptive signal control, achieving reductions in congestion of around 42% and emissions of around 35%. Uncertainty is handled through the preprocessing with Kalman filtering which accounts for measurement noise and therefore makes the predictions more accurate.

These papers show the shift of sensor fusion pipelines from statistical fusion to AI models and the shift toward real-time smart city systems. All three papers emphasize adaptive decision-making.
Common strengths in the papers are that multimodal sensing is seen as essential for reliable urban analytics, and the integrated AI/ML to enhance prediction and decision-making in every of the approaches. Weaknesses in the papers is the lack of cross-city validation since the models are not tested in diverse environments which makes it less applicable in the real world. The fusion pipelines are also not standardized which results in each paper using a unique architecture and there is limited discussion on interoperability between city services like policies, mobility, energy and environment.
Both Reis (2025) and Gantla et al. (2025) discuss the advantage of edge computing for low-

latency inference, reduced cloud dependency and resilience under network constraints which demonstrates a shift from centralized to distributed intelligence. Additionally, all three papers address sustainability but approach it from different angles. Mohanty et al. (2025) use data accuracy to save energy while Reis (2025) focuses on sustainable traffic optimization, and Gantla et al. (2025) use integrated adaptive control for improved environment management.

2.3 AIoT, CPSoS and Intelligent Urban Governance

AIoT-enabled Cyber-Physical-Systems of Systems (CPSoS) are redefining intelligent urban governance by fusing operational data streams with simulation-driven foresight. These systems combine short-term urban management with long-term strategic planning. Recent research highlights a shift toward generative, edge-enabled architectures that support sustainable, and adaptive urban governance.

Bibri and Huang (2025a) provide a conceptual and architectural analysis of the Urban Brain (UB) and the Urban DIgital Twin (UDT) systems, focusing on their integration through an AIoT-enabled CPSoS. They address one of the central challenges in intelligent urban governance: the missing connection between UB's real-time operational management and UDT's predictive and strategic simulation environment. One of their studies outcome is that while UB excels in stream-processed, operational intelligence, UDT provides scenario-based planning and predictive modeling but these systems rarely communicate effectively which leads to uninformed decisions, inefficiencies, and misalignment of operational and strategic urban goals.
The paper from Bibri and Huang (2025a) identifies information fusion as the missing piece between UB and UDT and proposes a framework where UB's real-time analytics continuously update UDT's simulation models and UDT's predictions inform UB's operational decisions. The proposed bidirectional feedback loop is designed to leverage computational capabilities and operational efficiencies. UB gets continuous IoT streams from distributed sensors, edge devices, and real-time CPS components. UDT combines heterogeneous, multi-source datasets for predictive modeling. AIoT is used to provide the computational backbone across and puts together cross-domain synchronization and systemic adaptability. Fusion occurs across multiple layers in the UB and UDT systems and the proposed framework emphasizes temporal fusion through uniting immediate operations in milliseconds or minutes with long-term planning in months or years from now.
Strengths of this paper are that it is one of the first papers that integrates the stream processing of UB and the simulation modeling of UDT. The framework is well-aligned with sustainability, resilience, and climate-oriented urban governance. Some weaknesses are that it lacks empirical validation or real-world tests.

In a related study, Bibri and Huang (2025b) write an extensive literature review and propose a novel framework for integrating Generative AI (GenAI) with IoT and AIoT across different domains of sustainable smart cities. Their literature review shows how Generative IoT (GIoT) enables synthetic data generation, imputation, scenario simulation, and autonomous learning which can help to improve the sustainability of smart city infrastructures. The paper identifies different ways of GenAI contributing to information fusion; the first one is multimodal fusion for comprehensive situational awareness where transformers support multimodal sequence fu-

sion across different IoT sensors and data. The second use case is generating cross-domain data fusion where embedded AI models work alongside generative architectures to create synthetic augmented data to enhance decision-making and adaptive control. Last but not least GIoT can extend traditional IoT by incorporating generative models that enable the synthesis of realistic data streams and autonomous behavioral patterns which helps to overcome data scarcity, class imbalance and sensor noise.
Bibri and Huang (2025b) identify different ways of uncertainty handling in their paper like VAEs that are highly interpretable and effective for uncertainty quantification or diffusion models that are useful for uncertainty-aware imputation tasks. GenAI has been established as a tool for producing realistic network traffic to overcome data scarcity, imbalance and representational limitations which is beneficial in complex urban systems with environmental uncertainty. In their paper they also mention that the environmental footprint of GenAI cannot be overlooked. Sustainable GIoT frameworks should consider environmental impacts from the beginning, balancing performance and intelligence with sustainability to support truly sustainable urban ecosystems. The paper clearly articulates how generative models solve key AIoT limitations like data scarcity, imbalanced data or missing sensor values and it provides a conceptual framework linking foundational technologies to system-level sustainability. A disadvantage is that it is very conceptual and lacks quantitative benchmarks on computational cost or energy efficiency tradeoffs. Additionally, the "sustainability paradox" in the form of GenAI's high energy footprint is acknowledged but not resolved with specific implementation pathways.

Gantla et al. (2025) present a multimodal fusion system which was previously discussed in chapter 2.2 and in this chapter we focus more on the Edge AI for low latency processing and blockchain for data integrity which can be seen in Figure 3. This system is designed for low-latency decision-making and secure, trustable data governance. Here fusion occurs through the Edge AI processing where the CNN, LSTM and RL algorithm run and on the Cloud AI & Big Data processing where data fusion and feature extraction is located. The blockchain handles trust uncertainty by ensuring secure data transactions and tamper-proof AI decisions which ensures the authenticity of sensor data. The system's ability to maintain 120ms decision latency introduces a form of fast decision-making critical for real-time governance of urban systems.
Gantla et al. (2025) offer an operational validation, unlike the more conceptual works of Bibri & Huang and demonstrate true multi-modal, cross-domain fusion. It also integrates AI, Edge computing, and blockchain into a deployable architecture with measurable improvements. A disadvantage is the focus on a narrow subset of city functions.

All papers recognize AIoT as a foundational technique that supports real-time data gathering, distributed processing, autonomous decision-making and system-wide optimization. All three papers emphasize the importance of sensor fusion and multi-modal data integration. The bidirectional feedback loop in the first paper makes use of UBs real-time streams and UDTs predictive simulation data at the same time. Generative AIoT (GAIoT) generates synthetic data to fill gaps and improve learning. Gantla et al. (2025)'s approach fuses traffic, pollution, weather, and noise in real time, all of these methods signal a shift from separated datasets to holistic, cross-domain data fusion.
The second Bibri and Huang (2025b) paper positions GenAI as a transformative force. In the other papers there are multiple opportunities to include GenAI to improve the frameworks such

as UDT's need for enhanced predictive simulation, UB's need for enriched real-time insight and Gantla et al. (2025)'s need for better training data. This indicates a paradigm shift from data-driven to data-generating smart city systems. All papers also position environmental sustainability as a principal driver. The UB–UDT integration supports climate-aligned planning and resource optimization, GAIoT introduces sustainability-aware generative modeling and federated learning and Gantla et al. (2025) optimize emissions, traffic, and energy through real-time fusion.

2.4 Energy Efficiency and Cybersecurity

Energy efficiency and cybersecurity are critical pillars of future IoT architectures, since smart city systems increase constantly in size, complexity, and autonomy. Recent research reflects a shift toward adaptive, hybrid architectures that optimize energy usage, data integrity, and system resilience in urban environments.

Alssaiari et al. (2025) address the dual challenge of energy consumption and anomaly detection for cyberattacks in IoT-driven smart city infrastructures through a hybrid deep learning black hole algorithm (DL-BHA) framework. Figure 4 shows this framework which integrates a LSTM network used for the modeling of energy consumption and anomaly detection with the Black Hole Algorithm (BHA) used for global optimization of control decisions. The paper highlights the difficulty of maintaining high-quality decision-making under resource constraints and data integrity risks, particularly for battery-powered and intermittently connected IoT nodes.
The DL-BHA framework performs information fusion through the multisensor data integration and the fusion of deep learning outputs and optimization decisions. The IoT dataset used with 17 characteristics, a temporal feature, a malicious indicator features, four IoT devices, and 11 smart-city application metrics represents a rich multisensor environment. The LSTM model processes the inputs through the normalization layer and uses the data to provide both energy consumption predictions based on regression and anomaly likelihood estimates based on classification. The Black Hole Algorithm operates on LSTM-derived features performing metaheuristic optimisation to find the best system-level control decisions. The integration enables dynamic threshold adaptation, allowing decision rules to change in response to fluctuating energy and anomaly patterns. The framework leverages the temporal pattern recognition capabilities of LSTM networks while exploiting the global optimization strengths of the black hole algorithm to analyse diverse data patterns from multisensor IoT systems. Alssaiari et al. (2025) set a new benchmark for dual-objective optimization in smart-city IoT systems with energy savings of up to 77.78% and anomaly detection accuracy exceeding 99%.
The strengths of this paper are the usage of multisensor temporal data which makes the research very valid and applicable in the real world and the outstanding performance of energy efficiency and anomaly detection. Additionally, does the paper attempt to address two problems at the same time that are usually treated separately which enables a completely new path in the research. A disadvantage is that it heavily relies on LSTM networks which are computationally expensive and therefore may not be feasible on low-power edge nodes and the optimisation process assumes relatively stable network conditions which ignores sensitivity to noisy environments.

Nassra and Capella (2025) focus on data reduction and compression to improve energy efficiency

in Wireless Sensor Networks (WSN) within smart-city deployments. They propose a hybrid data optimisation approach that integrates temporal grouping, difference-based data reduction, and adaptive Huffman encoding which can be seen in Figure 5. They are combining RMSSD-based similarity grouping used for reducing redundant transmissions and adaptive Huffman Encoding (AHE) for real-time compression resulting in a RMSSD-AHE algorithm.
Information fusion in this work is performed through compression-based and hierarchical fusion mechanisms: by grouping near-identical sensor readings based on the RMSSD threshold, the algorithm performs a form of temporal similarity fusion through merging sequences of repeated or near-repeated values into single entries. This improves the usefulness while preserving temporal order, which is essential for applications like structural or environmental monitoring. The AHE fuses symbol frequencies dynamically over time, learning an evolving coding structure without prior statistical assumptions. This real-time fusion of symbol statistics enables efficient compression even under non-stationary sensor behaviour in smart-city systems. Their layered 5G–IoT architecture performs functional fusion, distributing tasks across IoT devices, edge nodes, and cloud servers, balancing processing loads and integrating renewable-energy resources to support sustainability. The results show important improvements: a compression ratio of 3.55, a 71.85% reduction in power consumption, and extremely low reconstruction error of 0.01283%.
The method is computationally lightweight and therefore suitable for resource-constrained nodes and achieves strong performance which enables it to be used in real world applications. A weakness is that the compression may remove subtle data features important for other applications and that the method relies on the assumption that transmission cost is much higher than the processing cost, which may not be true for all IoT systems.

Both papers identify energy constraints in IoT devices as a primary barrier in smart city deployments. While Alssaiari et al. (2025) focus on resolving the trade-off between detection accuracy and energy use, the other paper indirectly addresses trade-offs between compression ratio and reconstruction possibilities. The literature demonstrates that sustainable smart city IoT systems require intelligent, adaptive, and multi-layered strategies that address energy efficiency, robust data handling, and scalable resource management. Together, these papers show a research trajectory in which IoT systems evolve from static, device-centric networks into dynamic, self-optimising, and resilient infrastructures suitable for the demands of modern smart cities.

3 Common Trends, Debates & Research Gaps across all topics

The majority of the papers focus on multisource data fusion as a core enabler for sustainable smart cities. It occurs on multiple levels and is a key tool in improving accuracy, robustness and decision making. There is a shift happening from statistical to AI based fusion where deep learning is used complementary to statistical methods to take advantage of both strengths. For the majority of the papers there is a common sentiment that urban intelligence must be uncertainty aware but just few of the papers actually implement uncertainty awareness in their frameworks. A consistent trend is to move towards edge AI and distributed processing to reduce latency and enable real-time governance especially for environmental monitoring and cybersecurity. Environmental sustainability is a central motivation in all topics and data fusion is seen as a mechanism to support sustainable planning of smart cities.

There is a tension between integrated uncertainty modeling methods that improve transparency and trust from urban planners but they are computationally expensive and methods without uncertainty awareness that scale better but provide weaker decision guarantees. Another debate is that deep models and AI achieve improved performance but at the cost of decreased explainability and edge feasibility. The sustainability paradox must also be recognized. GenAI and AI models are computationally and energetically costly and scalability remains challenging. The benefits of these technologies must be weighed against their environmental and operational cost. Another issue is that high predictive performance does not automatically result in governance trust or adoption due to the lack of explainability.

There are some research gaps when looking at the papers present in this literature review. Many of the research papers rely on simulations or single-city case studies for the validation and evaluation of their frameworks which do not represent reality. A lot of studies acknowledge real-world challenges but none fully demonstrate large-scale deployments. There is a lot of generalization and the differentiation between different urban contexts, climates, infrastructures and socio-economy settings is missing in most of the papers. There is also little work done on the integration between real-time operations and long-term planning deployed at scale which would be very beneficial for urban planners to make the best possible decisions. While energy efficiency is a recurring theme very few papers provide detailed benchmarks on computational cost and energy consumption of AI models. There is a lack of standardized sustainability metrics across AIoT fusion systems which makes it difficult to meet sustainability goals. Another research gap is robustness under distribution shift since urban systems and policies are constantly changing there is the need for transferability or domain-shift analyses which are not presented in the literature. Explainability in the smart city context is another topic that needs more research. Deep learning models offer accurate predictions at the cost of interpretability but planners and policymakers often require transparency to make trustworthy decisions. Last but not least the representation biases of the usage of synthetic data are acknowledged but rarely addressed within the fusion pipelines that use synthetic data.

4 Conclusion

This literature review shows that multisource data fusion, supported by deep learning, AIoT and distributed computing, is a key resource for the development and management of sustainable smart cities. The research papers show a clear shift toward uncertainty-aware, edge-enabled, data fusion frameworks that move cities from reactive management toward adaptive and real-time decision-making. While there has been significant progress in accuracy, efficiency, and scalability, current approaches appear to be small pieces of a bigger puzzle that do not connect together yet, with limited standardization and real-world validation.

4.1 Limitations

This literature review is based on only ten articles and therefore cannot provide a full coverage of the research in this field. Nevertheless, do the selected papers offer a representative overview of key themes and approaches. The scope of the review was further constrained by a ten-page limit, which restricted the depth of analysis. Last but not least, does the review place a narrow

emphasis on information fusion and uncertainty handling. Many of the reviewed papers address other relevant issues and solutions related to AI and ML as support for sustainable smart cities that could not be explored in detail.

4.2 Future Research

Future research should move beyond simulation-based evaluations and toward large-scale, real-world deployments that validate the multi-source data fusion frameworks across diverse contexts. Addressing robustness under distribution shift is critical since urban systems continuously change. Stronger integration between real-time operational systems and long-term planning tools is a challenge that future work must realize in actual deployment.

Another research direction is the development of uncertainty-aware sensor fusion frameworks that quantify and communicate uncertainty to make urban planning more risk-sensitive. The establishment of standardized sustainability metrics for AIoT fusion systems is another point that needs to be addressed in the near future. Research must also focus on energy-efficient model design and environmentally responsible usage of GenAI and deep learning.

Finally, future studies should prioritize explainability and fairness. Deep learning models and GenAI must be improved with interpretable mechanisms that expose the decision logic required for planning and policymaking. The representation biases also need to be identified, measured, and reduced within the fusion pipelines.

5 Finding & Accessing the papers

I verified the publication details for all of these research papers by clicking on the DOI and checking that the page it leads to is an actual research library. I also used professional databases like Google Scholar, MDPI and Science Direct to ensure the quality of the research papers used in this literature review.

Alssaiari et al., 2025: Balancing anomaly detection and energy efficiency in smart city IoT networks using hybrid deep learning and black hole algorithm
Found using the keywords "smart city IoT" on ScienceDirect. Focused on the hybrid deep learning + Black Hole Algorithm approach specifically in the beginning when data from different data sources is fused and the technology behind the energy efficiency. The information about this was taken from chapters 3 and 4.1.

Bibri and Huang, 2025a: Artificial intelligence of things for sustainable smart city brain and digital twin systems: Pioneering Environmental synergies between real-time management and predictive planning
Identified through ScienceDirect using "AIoT smart cities". I focused on the proposed framework improving the usage of Urban Brain and Urban Digital Twin. This information was mainly taken from chapters 5 and 6.

Bibri and Huang, 2025b: Generative AI of things for sustainable smart cities: Synergizing cognitive augmentation, resource efficiency, network traffic, cybersecurity, and anomaly detection for environmental performance
Identified through ScienceDirect using "AIoT smart cities". I focused on how generative AI is used in IoT systems for sustainable smart cities and on the proposed framework to further integrate GenAI with IoT in sustainable smart cities across functional domains. The information was gathered from the following chapters: 2.1.4, 2.1.6, 4.3 and 6.

Chithaluru et al., 2023: Energy-balanced neuro-fuzzy dynamic clustering scheme for green & sustainable IoT based smart cities
Found using the keywords "smart city IoT" on ScienceDirect. I focused on the proposed framework of an energy efficient dynamic clustering routing protocol using a neuro-fuzzy technique in order to restrict the resources of IoT devices. For this paper I focused on chapters 1, 3 and 4.

Gantla et al., 2025: Fusion of Real-Time Traffic and Environmental Sensor Data with Machine Learning for Optimizing Smart City Operations
Found through Google Scholar search using "sustainable AI based multi sensor fusion in smart cities". For this research paper I focused on the multi-modal fusion framework for urban traffic management and pollution control as well as the edge AI processing and blockchain technology to ensure secure data exchange. This information was found in chapter 3.

Jadhav et al., 2024: Spatiotemporal data fusion and deep learning for remote sensing-based sustainable urban planning
Found through Google Scholar search using "sustainable AI based multi sensor fusion in smart

cities". I focused on the methodology for spatiotemporal data fusion to integrate various urban data streams into pollution maps that are then used for simulations. Here I mainly focused on chapter 4.

Ma, 2025: Construction and uncertainty analysis of urban climate models under multi-source data fusion
Found through Google Scholar search using "sustainable smart cities multi sensor fusion uncertainty". I examined the proposed Adaptive Bayesian Hierarchical Multi-Source Fusion (ABHMF) framework with the integrated uncertainty propagation. When reading the paper I mainly focused on chapters 3 and 5.

Mohanty et al., 2025: Sensor Fusion Techniques for Comprehensive Environmental Monitoring in Smart Cities
Found through Google Scholar search using "sustainable AI based multi sensor fusion in smart cities". I primarily focused on the approach to use a Support Vector machine or a weighted average fusion model for real-time environmental monitoring of different data. For this paper I mainly took information from chapters 4 and 5.

Nassra and Capella, 2025: Hybrid big data optimization based energy-efficient and AI-powered green architecture toward smart cities and 5G-IoT applications Found using the keywords "smart city IoT" on ScienceDirect. I focused on the grouping process and on the overall functionality of the approach. I found the information in chapters 3.1, 3.2 and 6.

Reis, 2025: Internet of Things and Artificial Intelligence for Secure and Sustainable Green Mobility: A Multimodal Data Fusion Approach to Enhance Efficiency and Security
Found through Google Scholar search using "sustainable AI based multi sensor fusion in smart cities". For this paper I focused on the multimodal fusion approach for traffic optimization and the information for this was found in chapters 3 and 5.

References

Alssaiari, A., Alharby, M., Jan, Q., Hussain, S., & Ullah, S. (2025). Balancing anomaly detection and energy efficiency in smart city iot networks using hybrid deep learning and black hole algorithm. *Internet of Things*, *34*, 101800. https://doi.org/10.1016/j.iot.2025.101800

Bibri, S. E., & Huang, J. (2025a). Artificial intelligence of things for sustainable smart city brain and digital twin systems: Pioneering environmental synergies between real-time management and predictive planning. *Environmental Science and Ecotechnology*, *26*, 100591. https://doi.org/10.1016/j.ese.2025.100591

Bibri, S. E., & Huang, J. (2025b). Generative ai of things for sustainable smart cities: Synergizing cognitive augmentation, resource efficiency, network traffic, cybersecurity, and anomaly detection for environmental performance. *Sustainable Cities and Society*, *133*, 106826. https://doi.org/10.1016/j.scs.2025.106826

Chithaluru, P., Al-Turjman, F., Kumar, M., & Stephan, T. (2023). Energy-balanced neuro-fuzzy dynamic clustering scheme for green amp; sustainable iot based smart cities. *Sustainable Cities and Society*, *90*, 104366. https://doi.org/10.1016/j.scs.2022.104366

Gantla, H. R., Pandey, S. K., Mantha, S., Goyal, P., Jabeen, A., Fatima, S., & Mamodiya, U. (2025). Fusion of real-time traffic and environmental sensor data with machine learning for optimizing smart city operations. *Fusion: Practice and Applications*, *19*(2), 328–340. https://doi.org/10.54216/fpa.190224

Jadhav, S., Durairaj, M., Reenadevi, R., Subbulakshmi, R., Gupta, V., & Ramesh, J. V. N. (2024). Spatiotemporal data fusion and deep learning for remote sensing-based sustainable urban planning. *International Journal of System Assurance Engineering and Management*. https://doi.org/10.1007/s13198-024-02583-6

Ma, S. (2025). Construction and uncertainty analysis of urban climate models under multi-source data fusion. *International Journal of GEOMATE*, *29*(134). https://doi.org/10.21660/2025.134.5161

Mohanty, A., Mohapatra, A. G., Mohanty, S. K., Mohanty, A., Yang, T., Singh Rathore, R., & Wang, L. (2025). Sensor fusion techniques for comprehensive environmental monitoring in smart cities. *2025 3rd International Conference on Data Science and Information System (ICDSIS)*, 1–6. https://doi.org/10.1109/icdsis65355.2025.11070951

Nassra, I., & Capella, J. V. (2025). Hybrid big data optimization based energy-efficient and ai-powered green architecture toward smart cities and 5g-iot applications. *Journal of Electronic Science and Technology*, *23*(4), 100328. https://doi.org/10.1016/j.jnlest.2025.100328

Reis, M. J. C. S. (2025). Internet of things and artificial intelligence for secure and sustainable green mobility: A multimodal data fusion approach to enhance efficiency and security. *Multimodal Technologies and Interaction*, *9*(5), 39. https://doi.org/10.3390/mti9050039

Appendix

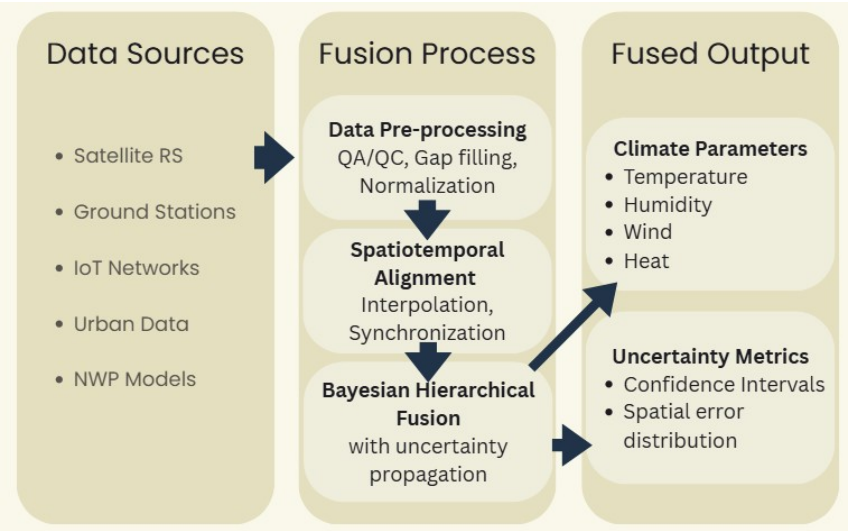

Figure 1: Multi-source data fusion framework - ABHMF (Ma, 2025)

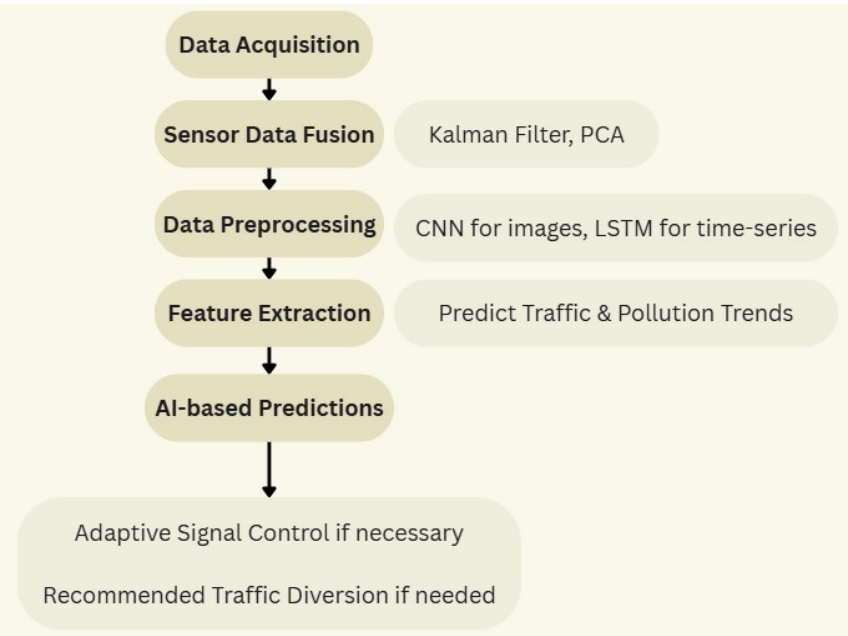

Figure 2: AI-driven optimization framework - Part 1 (Gantla et al., 2025)

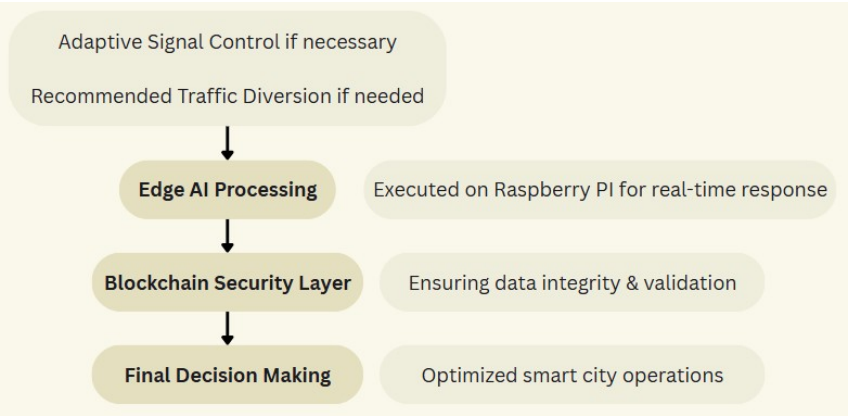

Figure 3: AI-driven optimization framework - Part 2 (Gantla et al., 2025)

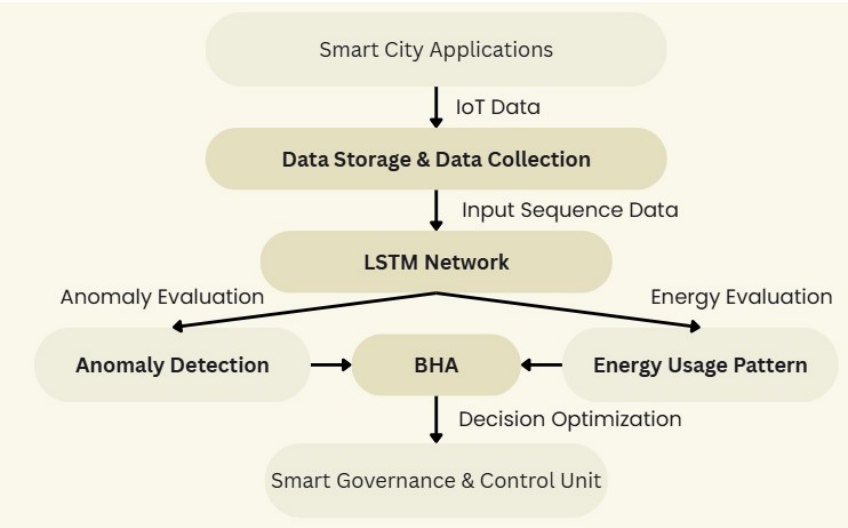

Figure 4: DL-BHA framework (Alssaiari et al., 2025)

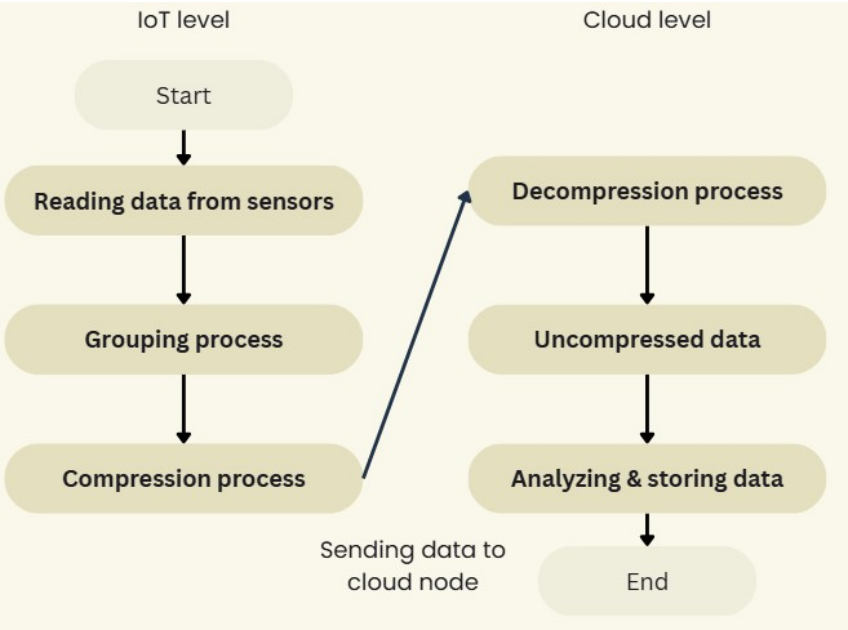

Figure 5: Proposed framework from Nassra and Capella (2025)